6 A pair of concentric coils is shown in Figure 6.1.

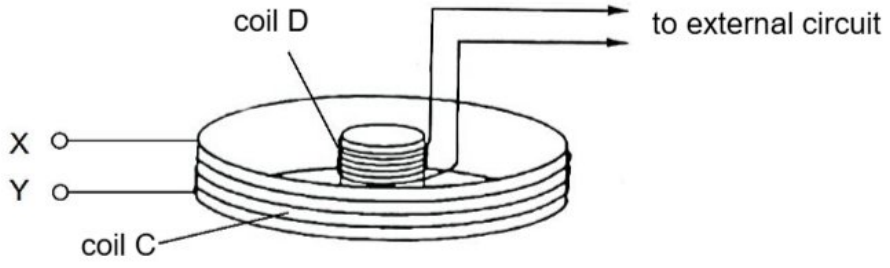


Figure 6.1

The outer coil C is connected to a variable power supply by the terminals XY.

The variation with time t of the current I in coil C is shown in Figure 6.2.

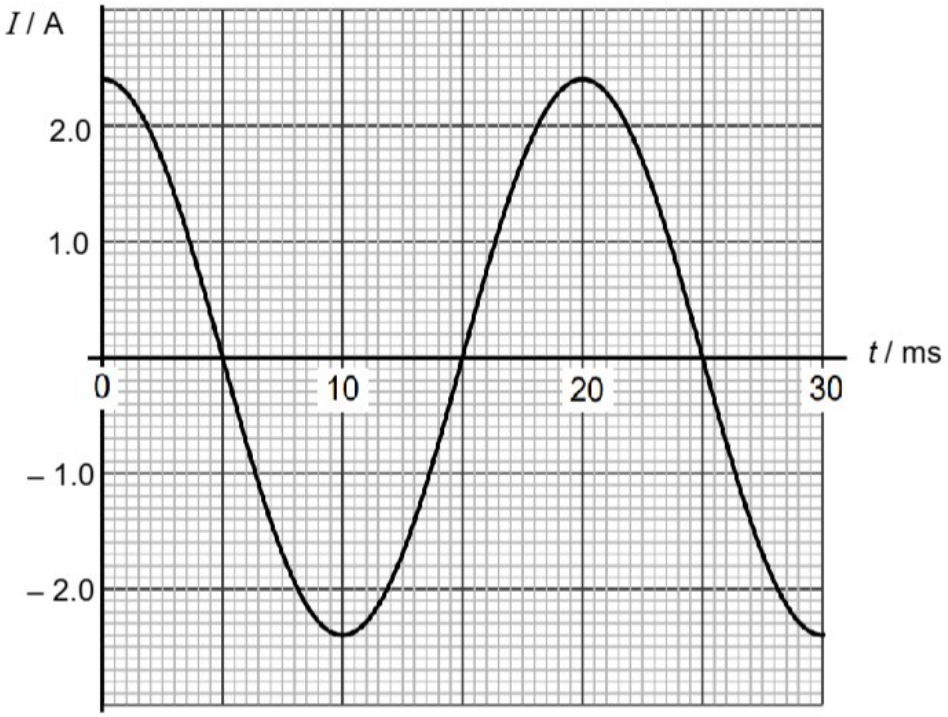


Figure 6.2

(a) Use Faraday's law to explain why an electromotive force (e.m.f.) is induced in coil D.

.....

.....

.....

.....

.....

..... [2]

(b) State an instant when the induced e.m.f. in coil D is a maximum value.

t = ...ms [1]

(c) Explain your answer for **6(b)**.

.....

.....

..... [1]

(d) The outer coil C produces a uniform magnetic field of flux density B, in tesla, in the region of coil D given by the expression

$$B = 1.30 \times 10^{-3} I$$

The coil D has a circular area of 1.05×10^{-3} m^2 and contains 50 turns of wire.

From t = 10 ms to t = 15 ms,

(i) calculate the average induced e.m.f in coil D,

e.m.f. = ...V [3]

(ii) use Lenz's Law to explain why the induced current in coil D is in the same direction as the current in coil C.

.....

.....

.....

.....

.....[2]

[Total: 9]